LAUREATE: A Dataset for Supporting Research in Affective Computing and Human Memory Augmentation

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The latest developments in wearable sensors have resulted in a wide range of devices available to consumers, allowing users to monitor and improve their physical activity, sleep patterns, cognitive load, and stress levels. However, the lack of out-of-the-lab labelled data hinders the development of advanced machine learning models for predicting affective states. Furthermore, to the best of our knowledge, there are no publicly available datasets in the area of Human Memory Augmentation. This paper presents a dataset we collected during a 13-week study in a university setting. The dataset, named LAUREATE, contains the physiological data of 42 students during 26 classes (including exams), daily self-reports asking the students about their lifestyle habits (e.g. studying hours, physical activity, and sleep quality) and their performance across multiple examinations. In addition to the raw data, we provide expert features from the physiological data, and baseline machine learning models for estimating self-reported affect, models for recognising classes vs breaks, and models for user identification. Besides the use cases presented in this paper, among which Human Memory Augmentation, the dataset represents a rich resource for the UbiComp community in various domains, including affect recognition, behaviour modelling, user privacy, and activity and context recognition.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in ubiquitous and mobile computing**; • **Applied computing** → *Education.*

Additional Key Words and Phrases: datasets, affective computing, human memory augmentation, student performance, physiological signals, learning, affect

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1 INTRODUCTION

In recent years, the development of wearable sensors has advanced rapidly, resulting in a wide range of devices available to consumers. These sensors have become smaller, more sophisticated, and consume less energy, enabling manufacturers to upgrade existing products and create new categories. However, hardware improvements alone cannot explain the success of wearable devices. The advances in the software domain, particularly in artificial intelligence (AI) and Machine Learning (ML), have also played a role in the massification of wearable sensors. Users of these ML-powered devices now have ways to monitor and improve their lives. Physiological signals captured by wearable devices enable users to analyse their physical activity, sleep patterns, cognitive load, and stress levels [5, 78, 90, 95, 106].

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The field of *affective computing* leverages wearable technology and AI to track factors associated with affective states (e.g. emotions, mental stress and cognitive load). While the area has seen significant advancements from the latest updates in hardware and software, labelled data is still one of the main bottlenecks [121] that impede the development of advanced ML models —models that can have significant real-life impact, including areas like digital healthcare [23]. Nevertheless, researchers must laboriously collect datasets to train the models that predict affective states, which is a process that presents several challenges.

First, despite technological improvements, the more powerful wearable sensors may still be cumbersome. Sometimes they are intrusive, expensive, or impractical (e.g. with short battery life or requiring frequent data downloading) [3]. The inter- and intra-subject variability of physiological signals present an additional challenge. Datasets need to contain data from multiple individuals across several sessions to achieve robust predictive generalisability, and recruiting volunteers with such constraints is demanding. Finally, some physiological sensors (such as galvanic skin response sensors) are highly sensitive and suffer from movement artefacts, variations in room temperature, and more [12]. These conditions limit data collection settings to controlled environments to maximise data quality. All these challenges make collecting in-the-wild datasets difficult and expensive. Therefore, researchers may be hesitant to share the collected data due to the required effort and financial expenditure.

A specific area of interest in this study are Human Memory Augmentation (HMA) systems. Affective computing methods can monitor mental states that are relevant to HMA. For example, physiological sensors have been used for cognitive load estimation [51], student engagement assessment [26], stress measurement [46] and more [35, 55, 113]. Since these mental states impact memory formation and retrieval [10, 14, 81, 118], various physiological signals have also been employed in human memory research [20, 50, 75, 96, 106].

In a university setting, large differences in the measured cognitive load among students may indicate important moments in the classroom for an HMA system. Students with high cognitive load may suggest that they find the content important but challenging. In contrast, students with low cognitive load at the same moment may indicate a lack of focus or loss of interest after struggling with the content. An HMA system could use contextual physiological information to identify relevant moments in a class or when a student was inattentive. Subsequently, it could assist the students in reviewing the corresponding information later.

This paper presents a dataset we collected during a 13-week study in a university setting. The dataset contains the physiological data of 42 students and 2 professors (i.e. lecturers) during 26 classes (including exams), as well as daily self-reports asking the students about their lifestyle habits (study, physical activity, sleep quality). The dataset includes participants' performance in quizzes (scores and answers to questions), assignments, and exams, among other data. The dataset is called LAUREATE: the Longitudinal multimodAl stUdent expeRience dataset for AffecT and memory rEsearch. It is available to researchers after the signature of a sharing agreement[1]. A snippet of the dataset has already been presented at UbiComp 2022 workshops [36, 67]. In contrast to the initial analyses, this study offers several additions, including: (i) access to the overall dataset; (ii) a detailed description of the data collection protocols; (iii) an exploratory data analysis for the complete dataset; (iv) access to expert features extracted from the raw data, along with an explanation of the pre-processing and feature extraction steps; (v) access to baseline ML models; and (vi) an analysis and discussion of future use-cases in affective computing and HMA.

The baseline ML models investigated in this study are models for estimating self-reported affect and mood-related states (understanding, satisfaction, energy, stress, tiredness, peacefulness, motivation, calmness, enthusiasm, and worriedness) using physiological data and behavioural self-reports (e.g. daily sleep quality and studying hours). We also investigated models that can recognise classes vs breaks using physiological and acceleration data. Such models would be useful for automatic activity and context recognition. And finally, we investigated models for user and device identification using physiological and acceleration data, which in the context of this work

[1]Please contact the first author for more details

could be helpful for unsupervised user adaptation. In a broader context, data that enables the development of user-identification methods are useful for improving user privacy mechanisms, another vital aspect of ubiquitous sensing systems [65].

In the following sections, we will (i) provide a detailed overview of the related work in publicly available datasets and the main differences with the LAUREATE dataset (Section 2), (ii) describe in detail the methods utilised for data collection —including participants' recruitment, experimental apparatus and procedure— and feature extraction (Section 3), (iii) apply statistical analysis and baseline machine learning techniques to validate the datasets (Section 4), and (iv) discuss the limitations of the dataset, its potential for HMA applications, future work and other possibilities (Section 5).

2 RELATED WORK

This section compares our dataset with publicly available datasets from five related research categories: emotions, mental stress and cognitive load, driver-state monitoring, workplace stress, and studies performed on student campuses. The overview tables contain information about the release date of the dataset, the number of participants, the duration of the study (e.g. one day represents that all data was collected in one day/session), the sensor data available in the dataset, and the experimental scenario which also hints the available ground truth (i.e. labelled) data. Regarding the sensor data, we focus on datasets that provide data from wearable physiological sensors such as electroencephalogram (EEG), electrooculogram (EOG), electrocardiogram (ECG), electromyogram (EMG), blood volume pulse (BVP), electrodermal activity (EDA), respiration rate (RESP), eye tracker, magnetoencephalography (MEG), skin temperature (TEMP), acceleration (ACC), beat-to-beat intervals (RR sensor), and pulse oximetry (SpO2) sensors.

In the two final subsections, we (i) discuss related research on human memory (Section 2.6), affect and cognitive load, and (ii) present the main differences between our dataset and the past work (Section 2.7).

2.1 Emotion Datasets

Emotion recognition from physiological data is a task that has produced many open datasets (Table 1). These are datasets where the participants have watched affective multimedia in short sessions and rated their experience after each session using questionnaires that provide information about the participants' emotional states (e.g. discrete emotions, arousal, and valence). The affective multimedia in each of these datasets consists of short movie or music video snippets (e.g. 1-minute duration) that are intended to elicit particular affective states. Various devices have been used to record the participants' physiological responses while watching the affective multimedia. These research and associated databases are helpful for analysing the emotional responses of participants in controlled settings.

2.2 Mental Stress and Cognitive Load Datasets

Monitoring mental stress and cognitive load is another category where researchers have been quite active, as seen in Table 2. In these studies, researchers typically expose participants to cognitive tasks, e.g. N-back task, Stroop test, Trier social stress test, or even smartphone games. While the participants are performing the specific tasks, their physiological response is recorded using various devices. Some of these studies combine emotion-related tasks (watching a set of funny video clips) and cognitive load tasks (e.g. WESAD [99] and Non-EEG datasets [8]). These studies and corresponding datasets are beneficial for studying the participants' physiological responses to mental stress and cognitive load tasks.

Table 1. List of available emotion datasets.

Dataset	Year	S.S.	Span	Sensor data	Scenario
DEAP [63]	2011	32	1 day	ECG, EDA, EEG, EMG, EOG, RESP, TEMP, face video	Valence, arousal, liking, dominance, and familiarity
MAHNOB-HCI[7]	2012	27	1 day	ECG, EDA EEG, RESP, TEMP, video, eye tracker, audio	Valence, arousal, dominance, predictability, and discrete emotions
RECOLA [89]	2013	46	1 day	Speech, Face, ECG, EDA	Valence, arousal, agreement, dominance, engagement, and performance
DECAF [1]	2015	32	1 day	ECG, EMG, EOG, MEG, near-infrared video	Valence, arousal, and dominance
SEED[129]	2015	32	1 day	EEG	Discrete emotions
MMSE [128]	2016	140	1 day	ACC 3D model sequences, videos, thermal videos, BVP, EDA, and HR	Discrete emotions
Ascertain[111]	2017	58	1 day	ECG, EDA, EEG, facial activation units	Big-5, valence, arousal, liking, engagement, familiarity
Amigos [77]	2017	40	1 day	EEG, ECG, GSR, face video	Valence, arousal, control, familiarity, liking, and discrete emotions
EmgDataVR [44]	2022	38	1 day	Facial EMG	Valence, arousal, and discrete emotions

S.S. stands for sample size.

Table 2. List of available mental stress/load datasets.

Dataset	Year	S.S.	Span	Sensor data	Scenario
Non-EEG [8]	2016	20	1 day	ACC, EDA, HR, TEMP, SpO2	Four stress types (physical, emotional, cognitive, none)
Mental Stress [46]	2017	21	1 day	ACC, EDA, HR, TEMP, BVP	Three stress levels (low, medium, high)
WESAD [99]	2018	15	1 day	ACC, EDA, TEMP, BVP, EMG, RESP	Neutral, amusement, stress
CLAS [76]	2019	62	1 day	ECG, EDA, PPG, ACC	Mental stress, Cognitive load, Arousal and valence
Cognitive load [45]	2020	23	1 day	ACC, EDA, TEMP, RR	8 cognitive load tasks with varying difficulty levels. NASA-TLX
Cognitive load [45]	2020	23	1 day	ACC, EDA, TEMP, RR	Smartphone game with three difficulty levels. NASA-TLX
Stress-Predict [59]	2020	35	1 day	BVP	Stroop test, TSST, and Hyperventilation

S.S. stands for sample size.

2.3 Driving Datasets

Another research category that produces publicly available datasets is the field of monitoring participants while they drive. Table 3 presents the studies that have produced publicly available datasets. The driving setup can be in a real-life scenario or a simulator. These studies provide valuable data points for developing future smart-driving services (e.g. driver attention management).

Table 3. List of available driving datasets.

Dataset	Year	S.S.	Span	Sensor data	Scenario
Driving-stress [54]	2005	24	1 day	ECG, EDA, EMG, RESP	Three stress levels (low, medium, high)
Driving-workload [100]	2013	10	1 day	GSR, HR, TEMP	Perceived cognitive load while driving
Driving-distractions [82]	2016	64	1 day	EDA,HR, RESP, facial expressions, eye tracking	Stress binary + NASA TLX

S.S. stands for sample size.

2.4 Workplace Stress Datasets

Table 4. List of available workplace stress datasets.

Dataset	Year	S.S.	Span	Sensor data	Scenario
TILES-2018 [79]	2020	212	10 weeks	PPG, ECG, Audio, ACC, TEMP	Hospital workplace. Personality, behavioural states, job performance, and well-being
TILES-2019 [125]	2022	57	3 weeks	PPG, ECG, Audio, ACC, TEMP	Hospital workplace. Personality, behavioural states, job performance, and well-being
Nurses stress	2022	15	Longitudinal (1250 h total)	ACC, EDA, TEMP, BVP	Nurses from an Emergency Room department.

S.S. stands for sample size.

Workplace stress is another category of research that has provided valuable datasets. Table 4 enumerates these works. Compared to the rest of the datasets, these are recorded longitudinally (e.g. at least three weeks) and entirely in the wild. Two datasets, TILES-2018 and 2019, provide access to behavioural and physiological data from a relatively large number of participants through IoT data hubs, wearable sensors, and a variety of surveys, including personality traits, behavioural states, job performance, and well-being. Similarly, the Nurses' stress dataset provides access to physiological and context data of nurses monitored during the COVID-19 outbreak. The context data includes information about the stress events (e.g. the period of the stress events and corresponding circumstances).

2.5 Student Campus Datasets

The studies that are mostly related to ours are StudentLife and Laughter study. In the Laughter study, Di Lascio et al. also measured students' and lecturers' physiological data. Although this dataset has a higher diversity of courses (five instead of two), their experiment only lasted three weeks, while ours lasted 13 weeks.

Table 5. List of available student campus datasets.

Dataset	Year	S.S.	Span	Sensor data	Scenario
StudentLife [120]	2014	48	10 week	Phone usage Students performance	University setting
Laughter [27]	2019	34	3 weeks	ACC, EDA, TEMP, BVP	Participants watched funny videos
Ours	2023	42 + 2	13 weeks	ACC, EDA, TEMP, BVP, Self-Reports Audiovisual Class recordings quizzes, class contents, performance	University setting

S.S. stands for sample size.

In StudentLife, 48 students from one class answered daily and weekly surveys for ten weeks about their lifestyle, mental well-being, and academic performance. The dataset also comprises sensor data from smartphones. Nonetheless, our dataset differs from it by adding physiological data from students and lecturers.

2.6 Affect, Cognitive Load and Human Memory

Harvey et al.[53] identified three key features of human memory: (i) "recall depends upon the relative distinctiveness of the target event when compared to its competitors", (ii) "human memory is cue driven" [112], and (iii) "cues captured by lifelogging technology can be used to augment human memory". A memory cue is something (an image, emotion, verbal or non-verbal sound, location, object, etc.) that can jog our memory, helping us recall the original experience. If a system presents to the user an experience — or its corresponding memory cue — at regular intervals, the user will repeatedly recall the experience through time and consolidate its memory, improving long-term retention [57, 91]. With sufficient practice, the user would only need to remember the specific cue to retrieve the full memory of the original experience.

Current capture technology could record full experiences and save them into long-term storage, working as a memory prosthetic. These systems would impose additional workload for its users, now forced to filter and select relevant memories. Human Memory Augmentation systems should focus on "selectivity, not total capture" [104]. We believe that physiological signals could be used as input to HMA systems for this task.

Mental states and memory are intertwined. Arousal stories are better remembered than neutral stories [14]; negative words [62] and experiences [61] as well. But not only the content's affect is important: the affective state of individuals is also related to posterior recall rates [108]. Cognitive load also relates to memory: divided attention (i.e. high cognitive load, with decreased working memory capacity) at encoding time negatively impacts recall scores [18, 21].

The UbiComp community has worked hard the past few years in affective state recognition [40, 41, 55, 92, 113] and cognitive load detection [47, 98, 106, 123] in diverse environments (including educational [37]) and with different objectives (such as HMA [20]).

We believe that these works can be applied to students in a University setting with a new objective: human memory augmentation. By detecting relevants moments of a class where students' physiology is correlated with low posterior recall, an HMA system could extract and create different memory cues (in the form of recorded class material: lecture slides, audiovisual snippets or flash-cards with class concepts) and present them to its users.

2.7 The LAUREATE Dataset

Our dataset should offer a valuable contribution to the UbiComp community, as it differs in several aspects from the previously detailed works.

First, to the best of our knowledge, there are no other publicly available datasets designed for Human Memory Augmentation. We only found two datasets analysing intracranial EEG (iEEG) of epilepsy patients during memory tasks [9, 34]. These datasets cannot be compared with ours, as the data modalities, participants, individuals and dataset purpose are different. Collecting datasets is always a demanding task, especially for in-the-wild settings. Additionally, the long-term capture of physiological signals through wearable devices introduces significant overhead for both experiments and participants. These factors may contribute to the limited availability of similar datasets in this area.

Second, all of the 19 datasets in the domains for monitoring emotions, mental stress and cognitive load, and driver state contain only one session per subject (see Table 1, Table 2, and Table 3), meaning that each participant has been recorded only on a single day. On the other hand, our dataset presents a longitudinal study of 42 participants across two courses during a whole university semester (13 weeks). This amount of data is a distinctive feature from other publicly available datasets, as it presents the possibility of analysing inter-subject and intra-subject correlations in the physiological data of the participants. Furthermore, thanks to the self-reports, it is also possible to analyse students' affective trends and study and lifestyle habits during the semester.

Compared to other in-the-wild datasets, mainly the *workplace stress datasets* (Section 2.4), our dataset benefits from the static essence of students attending lectures. This characteristic reduces the possibility of signal artefacts and noise in the data. In terms of data modalities, our dataset contains similar data to most of the previously listed datasets. Nevertheless, the difference from the primary dataset in the area, StudentLife [120], is the inclusion of students' physiological data.

Finally, different from all the existing datasets, our dataset includes students' performance on several in-class quizzes (both answers and grades), and timestamped topics in the classes. Including this data is beneficial for studying human memory, making it possible to analyse students' physiological and mental states during a particular topic and their posterior performance on a related quiz.

3 METHOD

3.1 Participants

As the experiment involved the participation of human subjects, ethics delegates at our Faculty examined and authorised the experimental protocol.

The experiment was carried out in two different courses at our university. The courses were chosen by convenience sampling [73]. One course was at the Bachelor's level and the other at the Master's level, ensuring no participant overlap between the two courses (*course_01*, *course_02*).

On the first day of classes, the students received a brief presentation about the experiment. The presentation included an explanation of the experiment's objective and general details about their participation requirements. Specifically, they were asked to wear a wristband during classes, complete daily surveys, and fill out up pre- and post-study surveys.

The students who expressed interest in the experiment provided their contact information on a pre-sign-up sheet. Afterwards, we sent these students an email with the informed consent and allowed them several days to review it. We then scheduled small-group video calls with the students to explain the experiment with more detail. During the call, we covered which types of data we would collect (physiological data, self-reported surveys, answers to quizzes, quiz scores, and exam grades), how to properly wear the wristband device, and demonstrated the mobile application used to collect daily self-reports. Students were assured that their data would not be seen by their professors nor used for grading purposes.

The next time we met the students (i.e. participants) in the classroom, we asked them to sign the informed consent. The informed consent contained a standalone checkbox asking for the participants' agreement to anonymously share their data with other investigators for research purposes. After signing the informed consent, researchers provided the participants an anonymous identifier consisting of a letter and a digit.

All participants were compensated with a gift card of their choice. The baseline remuneration was CHF 50.- for a 10-week commitment to the study (i.e. 20 sessions). The final payment depended on the amount of collected data and could amount to a maximum of CHF 100.-

3.2 Experimental Apparatus

This subsection details the tools used to collect physiological data (Section 3.2.1), the initial survey and daily self-reports (Section 3.2.2), and the final survey (Section 3.2.3).

3.2.1 Physiological Data. The physiological data were recorded using the Empatica E4 wristband[2], a medical-grade wearable device. It comprises several sensors which measure four different physiological signals of the user: electrodermal activity (EDA), blood volume pulse (BVP), acceleration (ACC), and skin temperature (ST). The signals are measured at different sampling rates. The device also derives the inter-beat intervals (IBI) and average heart rate (HR) in a ten-second span from the BVP signal. Details are available in Table 6.

Table 6. List of physiological signals recorded by the Empatica E4.

Signal	Sampling Rate	Comments
ACC	32 Hz	Data in [-2g, 2g] range
BVP	64 Hz	Data from photoplethysmograph (PPG)
EDA	4 Hz	Unit in microsiemens (µS)
IBI	Intermittent	Time between two beats with 1/64 second resolution
TEMP	4 Hz	°C (Celsius)
HR	1 Hz	Average over the last 10 seconds

3.2.2 Initial Survey and Daily Self-Reports. For the collection of self-reported data, we utilised a service called LifeData. This service consists of a web-based backend accessible to researchers and a mobile application used for data collection (self-reports). Researchers can design and visualise their surveys on a web browser. Once the surveys are finalised and published, participants can download them to their phones as a bundle (known as a *lifepak*) via the mobile application. The app collects the participants' answers and synchronises them to LifeData's backend, making them accessible for researchers to download.

It is also possible to configure the administration of the surveys, i.e. how users start a session. Participants can either initiate a session voluntarily or receive a reminder notification to start a new session. We chose the latter option as we were interested in the participants' self-reports right after their lectures to reduce recall bias [103]. If a participant did not respond to the notification, additional reminders could be sent. We sent an additional notification 40 minutes after the first one and allowed a 12-hour window to start a session from the first notification. Despite this wide availability window for answering the surveys, the average time elapsed on lecture days from notification until a session started was 18 minutes. Therefore, the participants would likely still accurately recall the feelings and impressions elicited by the class they just attended.

The initial survey session was started as soon as the *lifepak* was downloaded to the participant's phone. The survey comprised several questions about the participants' lifestyle and study habits. In particular, we asked

[2] https://www.empatica.com/en-int/research/e4/

them about their usual breakfast, their preferred sports exercises [17], smoking habits [30], their mother tongue, their English level, their confidence regarding having the skills for the class, their usual studying environment [4], and their expected grade. The questions can be seen in more detail in Appendix B.

As the mobile application assigns a unique anonymous ID to identify each installation of the application on a mobile phone, the initial survey also asked the participants to enter the study ID given by the researchers to match them both.

The daily survey was scheduled every day at 12:15, when the morning class block ends at our university. This time was chosen to maximise the probability of participants being free to answer the survey. If the day was a lecture day for one of the two courses, then the surveys for those participants were scheduled at the end of that lecture. The daily survey consisted of questions about the participant's habits in the previous 24 hours. The questions inquired about their physical activity [17, 115], breakfast ingestion, caffeine intake [60, 71, 107], study time [86] and sleep quality [124]. If the survey was after a class, it also included the PANAVA-KS scale and questions about the student's engagement in class, their level of immersion, and the perceived engagement of the lecturer [93, 94]. The daily survey questions can be seen in more detail in Appendix C.

Lecturers also received a survey after class. The survey was shorter than the participants', with identical engagement and immersion questions, the PANAVA-KS scale, and questions about their level of preparedness and performance. The lecturers' survey questions can be seen in more detail in Appendix E.

PANAVA-KS [97] is a scale used to assess the respondent's affective states. It is a modified version of the *Positive and Negative Affect Schedule* (PANAS) [122] scale, designed for ecological momentarily assessment (EMA). PANAVA-KS consists of a 10-item scale with bipolar adjectives representing high and low activation states of affect. The full scale measures the three dimensions of the affective state: positive activation (PA), negative activation (NA), and valence (VA)[3]. Our survey included the ten items of PANAVA-KS, which were rated on a 5-point Likert scale with its extremes corresponding to positive and negative states of high and low activation (see Table 7).

Table 7. PANAVA-KS [101]. Affective state dimensions. Values range from 1 (Low) 5 (High).

Positive Activation (PA)		**Negative Activation (NA)**		**Valence (VA)**	
Low	*High*	*Low*	*High*	*Low*	*High*
Bored	Enthusiastic	Peaceful	Angry	Dissatisfied	Satisfied
Spiritless	Motivated	Calm	Nervous	Unhappy	Happy
Tired	Awake	Relaxed	Stressed		
Weak	Energised	Worriless	Worried		

The questions on participants' engagement, perceived engagement, and immersion were adapted from previous work by Di Lascio et al. [26]. In this case, a 5-point Likert scale was used, with 1 representing *not engaged* and 5 representing *extremely engaged*.

For technical reasons, a second version of the daily survey had to be implemented after ~20 days from the start of the experiment. We took advantage of this situation to include additional questions (mainly the engagement and immersion questions) and clarify text unclear to participants.

3.2.3 Final Surveys. The final surveys were administered to the participants with the Qualtrics platform[4]. It was done in two parts, before and after the final exam. The pre-exam final survey included the same questions as the initial survey to evaluate potential habit changes in the participants. Participants were also asked to answer

[3]KS, which stands for Kurzskala, is the German abbreviation of "Short scale"
[4]https://qualtrics.com

several questions regarding their relationship with the course material [52], if they felt benefits from studying with teammates [2, 70, 72], and their comfort talking to the professor [93]. These emotional factors in students have been shown in the literature to be related to academic performance. The pre-exam survey also included the big-five questionnaire [13, 48, 110], a widely-used taxonomy of personality traits on five dimensions.

While the pre-exam survey asked participants about their perceived understanding of the course's theoretical topics and their expected final exam grade, the post-exam survey asked them about their actual understanding of the course's theoretical topics. The reason for reiterating a modified version of the question is to discover if the participants' perceived knowledge (pre-exam) of the content matched their actual performance (post-exam). The questions can be seen in more detail in Appendix D

3.2.4 Additional Material. Additional material on the dataset includes:

- in-class quizzes (participants' answers and their grades)
- exam topics and participants' grades
- class slides
- timestamped topics (i.e. in which moments the topics were being taught)
- automatically generated class transcripts

Audiovisual material (i.e. lecture recordings) may also be available under a more strict sharing agreement.

3.3 Experimental Procedure

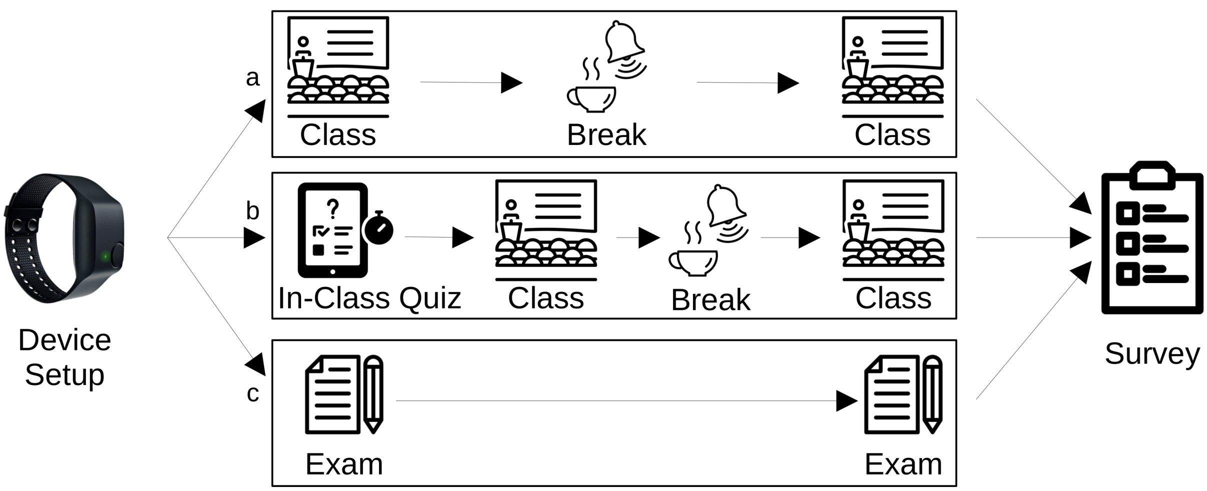
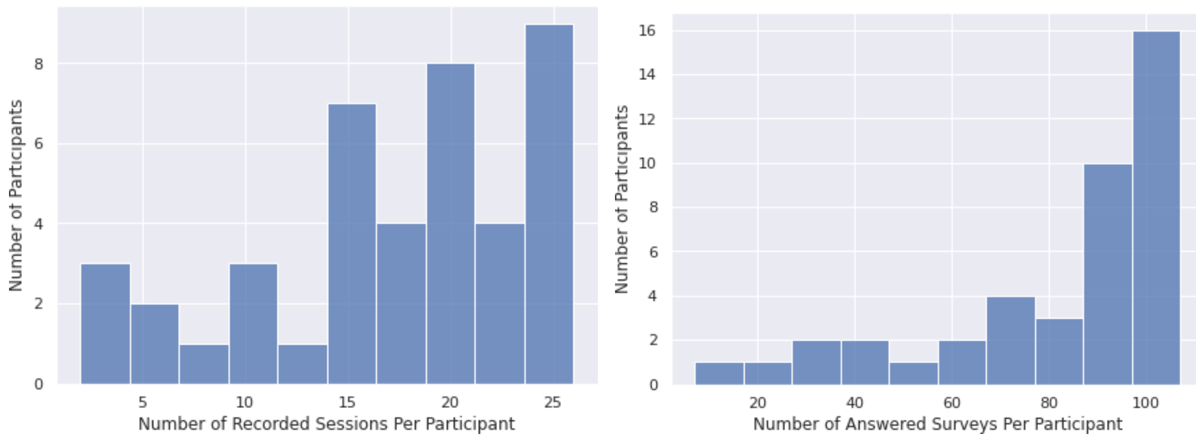


Fig. 1. Data collection procedure. Each session begins with the device setup. There are three types of sessions depending on their blocks (*class*, *quiz*, *exam*, and *break*): a) *class*-only sessions separated by a *break*, b) *class* sessions preceded by a *quiz*, and c) *exam* sessions. A survey followed all sessions.

The data-collection procedure is explained in Figure 1. We asked participants to arrive 10 minutes before the start of each class, to avoid delays and bottlenecks in setting up the devices. As students arrived at the classroom, we asked them for their study ID and noted which device was given to them. Before the start of the session, we equipped the participants with the E4 wristband on their non-dominant hand to minimise signal artefacts caused by movement while writing. As lecturers gesture more with their hands and walk while presenting, they used two devices —one device on each wrist to compensate.



(a) (b)

Fig. 2. Number of recorded sessions (Figure 2a) and answered surveys (Figure 2b) per participant.

Following Empatica's guidelines, we fitted the E4 above the wrist bone of the non-dominant hand, tight enough without generating discomfort in the user ("snugly above the wrist joint [...] but not so much as to cause discomfort" [5]). We verified that the PPG sensor lights were not visible without lifting the housing and asked participants if they felt any discomfort to avoid them manipulating the band. Once the device was secured in the participant's wrist, we turned it on and waited until its LED turned off, signalling the start of the recording.

The recording sessions could be of several types depending on the day's lecture. Some sessions were *class-only*, with *class* blocks separated by *breaks*. Other sessions started with a *quiz*, and were then followed by *class* and/or *break* blocks. *Exam* sessions consisted only of an *exam* block. *course_01*, additionally, had *lab* session types, consisting of hands-on sessions where participants interacted with each other while doing coding tasks.

After each session, the participants returned the devices and received the survey reminder on their phone (as explained in Section 3.2.2). We turned off the E4 devices and took them to the laboratory. We connected the devices to a single computer to charge them and synchronise them with Empatica's servers. This also enabled multiple-device synchronisation.

3.4 Duration, Environment and Dataset Size

The experiment was held during the *Spring Semester* of 2022, between February and June (northern hemisphere). Classes were held twice per week for each one of the courses, with no physiological data collected during the Easter break week. Classes ended at the beginning of June and final exams were held in mid-June, the official end of the experiment. Each course had 26 recording sessions.

The dataset contains data from 42 student volunteers (7 female, 35 male) and 2 lecturers (1 female, 1 male), i.e. the professor of each course. Participants averaged $\mu = 17.29$ recording sessions, with a standard deviation of $\delta = 6.62$. A histogram of the number of recorded sessions per participant is available in Figure 2a.

Daily surveys were administered from 2022-03-01[6] to 2022-06-17 (total = 109 days) for *course_01*, and from 2022-03-02 to 2022-06-14 (total = 105 days) for *course_01*. The average number of answered surveys per participant

[5] https://support.empatica.com/hc/en-us/articles/206374015-Wear-your-E4-wristband-
[6] Dates are expressed in the international format, i.e. $yyyy - mm - dd$

was μ = 83.62, with a standard deviation of δ = 26.09. A histogram of the number of surveys answered per participant is available in Figure 2b.

Additional dataset details are available in Table 8.

Table 8. Summary of the dataset's contents.

Data Modality	Data Amount
# participants	42 students (7f/35m), 2 lecturers (1f/1m)
# hours of physiological data	∼1400 hours (in 52 sessions)
# completed initial surveys	42
# completed daily surveys	∼ 3500 (∼700 on lecture days)
# quizzes	13 (6 for course_01, 7 for course_02)
# students answering quizzes	196 (105 for course_01, 91 for course_02)
# exams	3 (course_01: final exam; course_02: midterm and final exams)
# completed final surveys	34
# hours of audiovisual recordings	∼70 hours

3.5 Feature Extraction

This subsection details the feature extraction process from the different raw signals. The full list of features can be found in Appendix A.

3.5.1 Blood Volume Pulse. The Empatica device provides BVP data extracted from a PPG (Photoplethysmography) sensor for which they use a proprietary algorithm. The measurement unit is a fraction of nanoWatt (nW), representing the difference in light absorption observed by a light receiver in the PPG sensor. The frequency is 64 Hz. The BVP data is useful to quantify the dynamics of the heart rate, which in turn provides a limited view of the activation of the sympathetic and parasympathetic nervous systems. The sympathetic nervous system is responsible for speeding up certain processes within the body during a physically and/or a mentally demanding task (fight-or-flight response [19]). In contrast, after the demanding task, the activation of the sympathetic nervous system slows down, and the parasympathetic nervous system initiates the rest and repair processes.

In the context of our study, the sympathetic nervous system stimulation brought on by cognitive strain (e.g. lectures, quizzes or exams) also "stabilises" heart beats, resulting in more regular heartbeat intervals. The heart beat intervals, however, become less regular as a result of rest times in between jobs because "A healthy heart is not a metronome" [105]. The dynamics of the heart beats are frequently quantified using heart rate variability analysis (HRV). HRV analysis has been applied in a various studies for monitoring mental and physiological states, including emotions [127], stress [46], cognitive load [106], engagement [26], and sleep quality [38]. Similarly, our initial analysis of the dataset showed significant differences in the values of an HRV feature (SDNN) depending on the type of session the students were attending [67]. The students had highest SDNN values (most relaxed state) during standard classes. However, as expected, the SDNN values were lower during quiz and exam sessions (which represents increased stress levels).

To extract HRV features, we processed the BVP signal using the following steps:

(1) Segment the BVP signal on 2-minute segments with a 50% overlap between consecutive windows. The rest of the steps are applied for each 2-minute segment.
(2) Use a low-pass Butterworth filter of order three and a cutoff frequency of 5 Hz.
(3) Use a moving average filter of 16 samples (0.25 seconds).

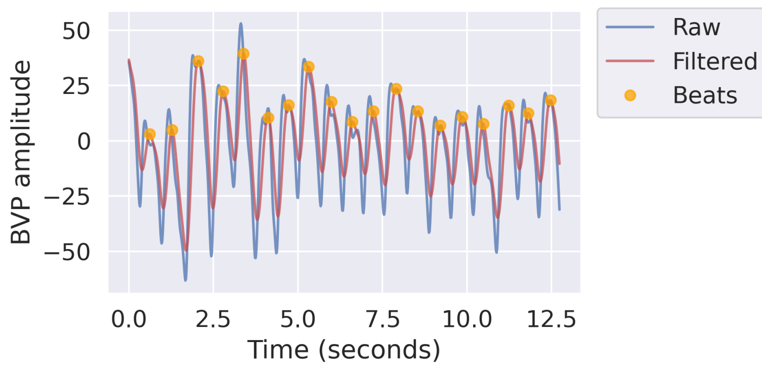


Fig. 3. Filtered BVP signal (red lines) and corresponding heartbeats detected automatically (orange dots).

(4) To detect the heartbeats, we used a peak detection algorithm based on the first-order difference of the filtered BVP signal [80]. The peak-height threshold was set to 75% of the median of the filtered BVP signal. The minimum distance between two consecutive peaks was set to 50% of the median peak distance for the given window. Example filtered BVP signal and detected heartbeats are depicted in Figure 3 (orange dots).

(5) The difference between two consecutive heartbeats represents the peak-to-peak (PP) intervals (also known as RR or NN intervals). We removed the PP intervals that were outside the range [.7 * median-PP, 1.3 * median-PP], where median-PP is the median duration of the PP intervals in the given 2-minute window.

From the PP intervals, we extracted 13 HRV features from the time- and frequency-domain. The PP intervals we extracted from the BVP signal correspond to the Inter-Bit-Interval (IBI) data provided by Empatica. However, Empatica provides IBI data only in low-movement periods. Table 9 compares the IBI data provided by Empatica, and the PP data we extracted from the BVP signal. From the table, we can see that less than 1% of the PP data is missing, which is a much lower percentage and thus better than the 36% of missing IBI data. To verify the quality of the PP data, the table also presents the Pearson's Correlation Coefficient (PCC) and the Mean Absolute Error (MAE) for two features (heart rate and SDNN) extracted from the PP data and from the IBI data. For the heart rate, the PCC is 0.95 (with a p-value $< .001$), showing an almost perfect match between the heart rate calculated from the PP data and the heart rate calculated from the IBI data. The MAE is 2.54 Beats Per Minute (BPMs) with a standard deviation of 3.76. For the SDNN, the PCC is 0.67 (with a p-value $< .001$), showing a strong correlation between the SDNN calculated from the PP data and the SDNN calculated from the IBI data. The MAE is 0.03 seconds with a standard deviation of 0.02 seconds. These metrics indicate that the quality of the PP data is at least as good as the quality of the Empatica data, especially when we look at averaged features such as the heart rate. The difference between the PP data and the IBI data is larger for the more time-sensitive features, such as the SDNN. Nevertheless, the percentage of missing PP data is much lower than that of missing IBI data.

Table 9. Comparison between the Peak-to-Peak (PP) data extracted from the raw BVP data and the IBI data provided by Empatica. The comparison includes the missing data percentage, and PCC and MAE across two features, heart rate (HR) and the standard deviation of the PP intervals (SDNN).

Data Source	Missing Data	HR PCC	HR MAE (BPMs)	SDNN PCC	SDNN MAE (seconds)
BVP (ours)	0.1%	0.95 (p<.001)	2.54 (3.76)	0.67 (p<.001)	0.03 (0.02)
IBI (Empatica)	36%				

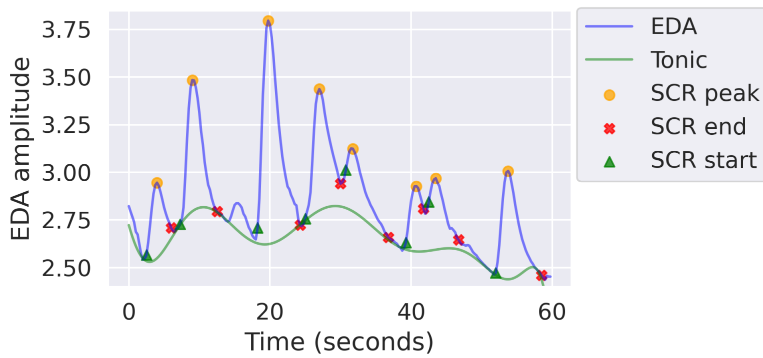


Fig. 4. EDA signal (blue line), Tonic component (green line) and Skin Conductance Reponses (SCRs). Each SCR is depicted with three corresponding points: start (green triangles), peak (orange dots), and end (red crosses).

3.5.2 Electrodermal Activity. Electrodermal Activity (EDA, or Galvanic Skin Response —GSR) refers to electrical changes that arise when the skin receives specific signals from the brain. In particular, EDA is described by physiologists as produced purely by the sympathetic nervous system, thus argued to represent a sensitive measure of the activation of the sympathetic nervous system [22, 85]. Consequently, EDA has been used in many studies to measure emotional activation, cognitive workload, physical exertion, etc. [11, 39, 116]. EDA can be measured using small electrodes attached to the skin. For example, the Empatica E4 wrist device measures EDA by flowing a minuscule amount of current between two dry electrodes, through which the device measures the conductivity on the wrist. The data unit is microsiemens (μS), and the frequency is 4 Hz.

Besides being a measure of average sweating levels, EDA is also characterised by Skin Conductance Responses (SCRs), which are caused by external stimuli. A typical SCR lasts 1–5 seconds, has a steep onset and an exponential decay or recovery, and reaches an amplitude of at least 0.1 μS [11, 117]. Consequently, SCRs can be used as an additional measure of stimuli presence and stimuli intensity.

Figure 4 presents an example EDA signal with the corresponding fast-acting component (SCRs) and slow-acting component (tonic signal). Each SCR is depicted with three corresponding points that are automatically detected: start (green triangles), peak (orange dots), and end (red crosses). We process the EDA signal using the following steps:

(1) Use polynomial curve-fitting on the EDA signal to decompose into two components (SCRs and tonic component).
(2) Subtract the tonic component from the EDA signal and use the resulting signal as an input to a peak detection algorithm.
(3) Detect SCR peaks using the first-order difference of the signal, with a height threshold of 0.1 μS. The minimum distance between two consecutive peaks was set to one second.
(4) For each SCR peak, detect surrounding valleys, i.e. the start and end of the SCR. The valleys are detected again using first-order difference.
(5) Remove SCRs that have: amplitude-increase duration shorter than one second; or amplitude-decreasing duration shorter than one second. For clarification, the amplitude-increase duration is the timespan between the start of an SCR and the peak of the SCR. Amplitude-decrease duration is the timespan between the peak of the SCR and the end of the SCR.

From the decomposed EDA signal, we extracted 41 expert features: 12 features from the raw EDA signal, 19 features related to the SCRs, and 10 features related to periods with significant EDA changes [49]. A significant change was defined as a 5% change in the EDA signal sustained for a period of 15 seconds.

3.5.3 Accelerometers. The Empatica device provides 3-axis accelerometer data represented by Earth's gravity across each of the three spatial dimensions (x, y, and z). The scale is limited to ±2g, and the frequency is 32 Hz. We processed the ACC signal using the following steps:

- Calculate the magnitude of the acceleration signal.
- Use a moving average filter with 8 samples (0.25 seconds).
- Compute the periodogram over the filtered magnitude.
- Extract features from the magnitude and from the periodogram.

3.5.4 Skin Temperature. The "fight-or-flight" response restricts blood flow from the extremities and enhances the vital organs. This peripheral vasoconstriction produces changes in the skin temperature of the extremities. As a result, skin temperature on the hands can indicate stress. Moreover, it can reveal the intensity of acute stress [56]. The Empatica device provides peripheral skin temperature data with a frequency of 4 Hz.

3.5.5 Statistical Features. In addition to the expert features, we extracted 22 statistical features from each of the four signals (BVP, EDA, ST, and ACC). The statistical features are 11 statistical descriptors applied to the filtered signals and to the first derivative of the filtered signals. The 11 statistical descriptors are: the first four statistical moments, the mean value of the first and second derivative, the 25th and 75th percentiles, the quartile deviation, the difference between the maximum and minimum values, and the coefficient of variation.

3.5.6 Inter-Subject Correlation. Recent studies have shown that, besides autonomic function (e.g. breathing), heart rate can also be modulated on attentional focus [66, 68, 83]. For example, Pérez et al. [83] showed a significant inter-subject correlation of heart rate when subjects are presented with an auditory or audiovisual narrative. Given that in our study participants are presented with the same audiovisual narrative, e.g. the lecture, inspecting inter-subject correlation may be a research question of interest.

We calculated four Inter-Subject Correlations (ISC) based on the Empatica signals, ISC-EDA, ISC-ACC, ISC-HR and ISC-ST. The ISCs were calculated using the following steps:

(1) For each joint session (e.g. a lecture), synchronise the data across the participants.
(2) Only for the HR data, upsample the PP (peak-to-peak) signals to a continuous sampling frequency of 4 Hz. The EDA and the ST data are already provided at 4 Hz. The ACC data were downsampled from 32 Hz to 8 Hz.
(3) For each signal, calculate ISC by following a similar approach to Pérez et al. [83]:
    - Segment the data to 1-minute windows.
    - Perform window-based z-score transformation for each 1-minute window.
    - Calculate pair-wise Pearson's correlation (i.e. each participant with each participant) for each 1-minute window.
    - Using the pair-wise correlations, calculate the average Pearson's correlation per subject (excluding self-correlation). Note that Pérez et al. [83] also apply the inverse of the fisher-Z transformation to the average PCC to calculate the final ISC scores, but we skipped this step so the final numbers in our dataset represent the average PCC correlations per subject.

4 VALIDATION

In this section, we will present validation results from the dataset. We will provide initial statistical analysis (Section 4.1) and preliminary ML experiments (Section 4.2).

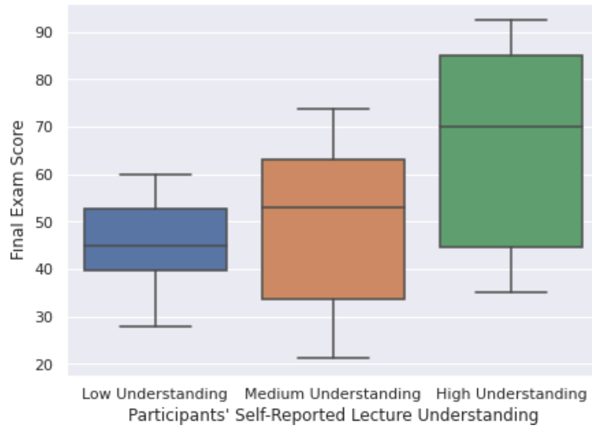


Fig. 5. Comparison of the distribution of final exam scores grouped by participants' self-reported lecture understanding. The mean value of understanding for the whole experiment was calculated for each participant, and then the participants were categorised according to that value into three bins of equal size.

4.1 Statistical Analysis

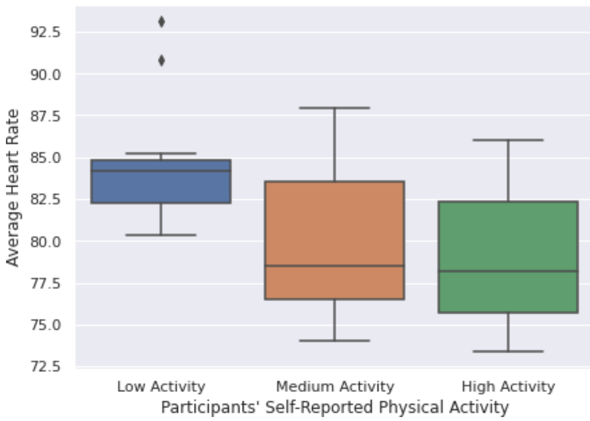
This section covers exploratory data analysis over the dataset. We will (i) analyse self-reported lecture understanding its relationship with final exam scores (Section 4.1.1), and (ii) analyse self-reported physical activity and its relationship with heart-rate features. Additional analyses exploring trends throughout the semester and the relationship between self-reported engagement and electrodermal activity features are available in Appendix F.

4.1.1 Reported Understanding and Final Exam Score. Knowing what one does not know is a crucial skill in learning [16, 69, 126]. If a student is aware of their knowledge gap, they are one step further in the path to remediate it. A possible question to answer with our dataset is if the students are aware of their understanding, or lack thereof, of the concepts taught during the lectures. We analysed the relationship between the students self-reported understanding of the lectures and their final exam score. The results are shown in Figure 5

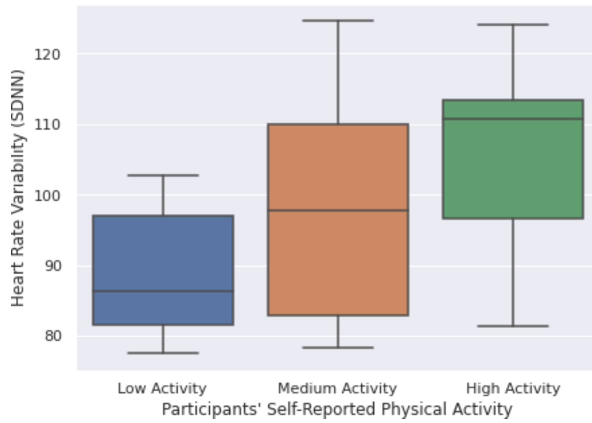
Figure 5, a boxplot, shows the differences in final exam scores of participants according to their level of understanding of the lectures. Participants answered to the question "How much of today's session have you understood?" (i.e. *lecture_understanding*, Appendix C). We then calculated each participant's mean *lecture_understanding* score, and then split the participants in three equally-sized groups using quantile-based discretisation with three bins. The image shows a tendency for groups "Low Understanding" and "High Understanding" to differ with respect to their final exam performance. Nevertheless, the results from a Kruskal-Wallis test [64] with post-hoc Dunn's analysis [32] fail to reject the null hypothesis (with a p-value = 0.079), probably due to a small sample size in each group. For this comparison we only considered participants that answered to at least 10 post-lecture surveys, leaving us with a sample size of $n = 36$ (i.e. 12 participants per group).

When conducting experiments that involve participants' self-reports, there is always a risk of bias that researchers must consider and try to minimise. In our analysis, we aimed to measure how students' self-reported understanding of the course contents, a measure subject to bias, matched their actual performance in the course, an objective measure of their comprehension. Even though we did not find statistically significant differences between the groups, the trend suggested that students in both the Low- and High-Understanding groups were able to assess their understanding of the course accurately, as reflected by their grades. This finding lends some validity to the self-reported data.

4.1.2 Physical Activity and Heart Rate Features. Our dataset offers the possibility of analysing the impact of students' self-reported lifestyle habits with their physiological signals. An example of this is the analysis of participants' self-reported physical activity and their physiological signals, namely their heart rate features. Figure 6 presents this analysis.



(a) Comparison of the distribution of average heart rate values grouped by participants' self-reported physical activity. The mean value of physical activity for the whole experiment was calculated for each participant, and then the participants were categorised according to that value into three bins of equal size. The average heart rate represents each participant's mean heart rate value over all recording sessions.



(b) Comparison of the distribution of mean heart rate variability grouped by participants' self-reported physical activity. The mean value of physical activity for the whole experiment was calculated for each participant, and then the participants were categorised according to that value into three bins of equal size. The heart rate variability represents each participant's mean heart rate variability over all recorded sessions.

Fig. 6. Comparison of heart rate features (Figure 6a, average heart rate; Figure 6b, heart rate variability —SDNN) between groups of low, medium and high self-reported physical activity

Figures 6a and 6b present participants grouped according to their answers to the question "Did you exercise since yesterday's survey?" (i.e. 10.3, *daily_activity*, Appendix C). The mean self-reported *daily_activity* value was calculated for each participant. The participants were then divided into three equal-sized groups with similar physical activity levels using quantile-based discretisation with three bins. Then, the three groups were compared with regard to their mean heart rate (Figure 6a) and mean heart rate variability (Figure 6b) across all lectures. The difference in the groups is statistically significant according to the Kruskal-Wallis test, rejecting the null hypothesis for both mean heart rate (with a p-value = 0.027) and mean heart rate variability (with a p-value = 0.015) comparisons. The post-hoc Dunn's test confirms that the difference is significant in both cases between the groups of low and high levels of physical activity.

The figures reproduce the results from previous works in the areas of physical fitness and cardiology: resting heart rate decreases with physical activity [88], while heart rate variability increases in physically active adults when compared to sedentary individuals [24, 31]. As in the previous section, this result adds some credibility to the self-reports from students, but also to the quality of the measured physiological signals. The values of the physiological signals reflect the self-reported fitness levels of the participants.

4.2 Machine Learning

This subsection presents baseline ML models we developed for:

- user- and device-identification using physiological and acceleration data. These would be useful for unsupervised user adaptation, but also as baseline models for evaluating user privacy. In these experiments, one instance represents a 2-minute data window, i.e. the models try to identify the user every two minutes. We experimented with classical ML models (trained using the features extracted from 2-minute data windows), and with Deep Learning (DL) models (trained using the raw data from the same 2-minute data windows).
- classes vs. breaks recognition using physiological and acceleration data. Similar to the user-identification models, also in these experiments one instance represents a 2-minute data window, i.e. the models try to detect whether the users are at class or are having a break every two minutes. We experimented with classical ML models, and with DL models.
- estimating self-reported affect and mood states (understanding, satisfaction, energy, stress, tiredness, peacefulness, motivation, calmness, enthusiasm, and worriedness) using physiological data and behaviour self-reports (e.g. sleep quality and study hours prior to a specific recording session). Different than the other type of models, in these experiments one instance represents one session (e.g. one class). That is because the self-reported data (labels) corresponds to the overall session. Because of the per-session labelled data, the number of instances was 671, which may not be enough for an end-to-end learning approach.

The classical models were trained using Random Forest (RF) and Light Gradient Boosting Machine (LGBM), algorithms that require minimal parameter tuning and produce models that are less sensitive to noise because of their ensemble-based approach. Although ensemble approaches are considered classical, they are still relevant in the ML community, producing state-of-the-art results in human activity recognition, for example [119].

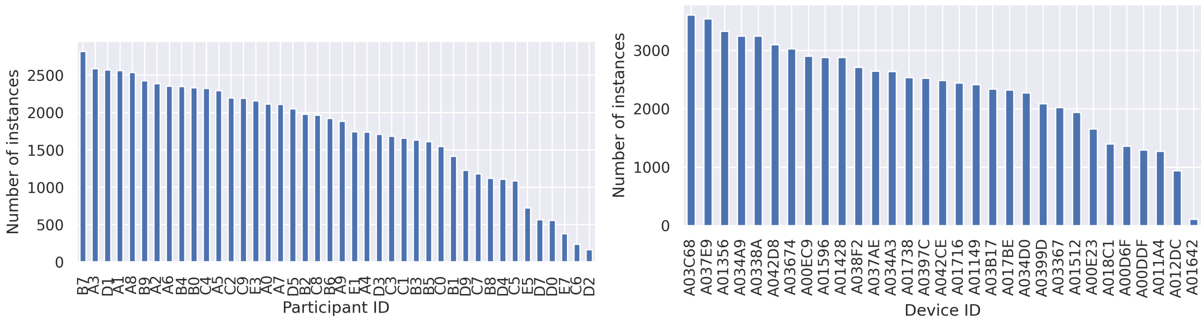
As an end-to-end DL approach, we used Spectro-Temporal Residual Network (STResNet). STResNet is an architecture for end-to-end learning specialised for multimodal sensor data. More specifically, STResNet can learn from several sensors simultaneously, each of them having a different sampling frequency, and it learns both in the time domain and in the frequency domain. The frequency information is extracted by calculating the spectrogram in decibels for each sensor channel; the temporal representation is extracted by residual blocks that contain CNN layers with 1-dimensional filters. The final output is provided by a softmax layer, which represents a class probability. STResNet was introduced for activity recognition based on smartphones [43], but later on, it has been used for a variety of domains including physiological sensors [109], and sound [42].

Our goal in these experiments was to provide baseline results for future research. We did not perform extensive hyperparameter tuning, feature selection, nor data normalisation steps. The features and the code for training the models will be made available online with the dataset.

4.2.1 User Identification and Device Identification Using Wrist-Device Data. User identification is a task that has been explored from several perspectives, including unsupervised personalisation, user privacy, and access control (security). For example, user identification based on wearables could replace the conventional password, enabling automatic personalisation of services and access control of confidential information. Example approaches based on sensor data include user identification using IMU data while performing daily activities [33], user identification using leakage current measured during laptop usage [29], and user identification using RFID data [58].

Our dataset is unique in regard to the domain of user identification, containing dozen 1.5 h sessions per participant scattered through four months period. Furthermore, the participants were assigned a random Empatica device for each session (one out of thirty-one devices), thus decoupling device-specific bias in the data from the user IDs.

The classical ML models were trained using the features extracted from 2-minute data windows, and the DL models were trained using the raw data from the same data windows. Based on the input sensor data, the models learn to detect the user that wears the device, i.e. the user that provides the input data. The number of windows (instances) per participant and per device after the segmentation are presented in Figure 7a and Figure 7b. For



(a) (b)

Fig. 7. Number of instances per participant (Figure 7a) and device (Figure 7b) after sliding-window segmentation

example, participant B7 has 2817 2-minute instances, which amounts to 1408 unique minutes because of the 50% overlap between consecutive windows (or 23 hours and 28 minutes of data). The average number of instances per participant was 1741. The average number of instances per device was 2359.

Figure 7a presents a skewed distribution due to some participants dropping out from the experiment, and sometimes even the course, soon after starting the semester. Attrition is a normal occurrence in universities, and one of the challenges in data collection. Figure 7b also presents a skewed distribution, but due to different reasons. Device **A01642** (first from the right) stopped working after the first recording session, so it was not used anymore. The reason why the following six devices from the right also have a lower number of data instances is because they were used for a parallel experiment. Due to the early drop in participants, there was a surplus of devices for our data collection. Older devices (i.e. with smaller IDs in lexicographic order) were given to the other experiment.

We evaluated the models using 5-fold cross-validation (without data shuffling), where all the data collected in one day belongs only in one of the folds, and consequently, it belongs either only in training or in test subsets. This evaluation approach avoids temporal overlap between the train and the test subsets. In a real-life use case, this would mean that the models have access to labelled sensor data from previous days (sessions), i.e. the models have labels about which sessions and consequently which 2-minute windows belong to which user. Based on the training data, the models try to identify (detect) the user to which the 2-minute windows (from the test set) belong. The results of these experiments are shown in Table 10. For the user-identification task, all of the ML models outperformed the majority classifier by a large margin. The highest performance of 56.3% accuracy and .49 F1-score was achieved by the STResNet model. These results indicate that users can be identified to some extent from the physiological data using ML and that end-to-end learning may be a more appropriate approach for the task. For the device-identification task, the results show that the ML models did not outperform the majority classifier by a large margin, which indicates that there is no substantial device-specific bias in the data.

4.2.2 Class vs. Breaks Recognition Using Wrist-Device Data. Class and breaks recognition from the Empatica data could be considered as a use-case for Human Activity Recognition (HAR) methods – a well-established research field that develops methods for recognising the gestures and motions a user performs. Recognising when the students are taking a class or on a break could lead to contextualised interventions like blocking distractions during work activities or suggesting the students take a break during long studying sessions. For example, Di Lascio et al. [25] proposed a multimodal approach – that includes Empatica data, laptop usage, and phone usage data – to automatically recognise the breaks and work activities of knowledge workers. In a study involving 14 participants, the ML model proposed by the researchers achieved an average F1-score of .82.

Table 10. Performance measures for user-identification and device-identification ML models. The models were evaluated using 5-fold cross-validation (without temporal shuffling).

Model	**User Identification**		**Device Identification**	
	Acc. (%)	F1-score	Acc. (%)	F1-score
Majority	3.8	N/A	4.9	N/A
RF	40.1	.34	8.9	.08
LGBM	27.3	.23	9.5	.09
STResNet	56.3	.49	9.1	.08

Similar to the user-identification experiments, the classical ML models were trained using the features extracted from each 2-minute window, and the DL models were trained using the raw data from the same windows. We evaluated the models using 5-fold cross-validation (without temporal shuffling), where all the data collected in one day belongs only in one of the folds, thus avoiding temporal overlap between the train and the test subsets. The results of these experiments are shown in Table 11. The table reports evaluation scores, for per-instance predictions and for voting predictions. Instance predictions treat each user as independent from the group, i.e. the evaluation scores are calculated over all the instances from the sliding-window segmentation. Grouped prediction utilises the information that several users are attending the same class, and thus the same break segment could be recognised using the data from different users. For example, if there are 15 students attending the same class, there will be 15 predictions for each 2-minute segment (instance). We grouped those 15 predictions using threshold-based voting: predict break if a break was recognised in 20% of the predictions (e.g. 3 out of 15). We chose 20% instead of 50% (which would be a majority vote), because we wanted to prioritise the "break" class, given that the class distribution was skewed towards the "no-break" class – 88% of the instance had a label "no break".

If we look at the evaluation scores before the voting (instance predictions in Table 11), we can see that all of the ML models performed similarly, with STResNet and LGBM having a slightly higher F1-score than RF. It can also be seen that the F1-scores are lower than the accuracy scores – a consequence of the class imbalance. The voting-based predictions brought significant improvements compared to the instance-based predictions, with the highest F1-score of .9 achieved by the STResNet model which is much higher than the F1-score of .72 achieved by the same model for the instance-based predictions.

Table 11. Performance measures for breaks-detection ML models. The models were evaluated using 5-fold cross-validation (without temporal shuffling).

Model	**Breaks detection (instance predictions)**		**Breaks detection (voting predictions)**	
	Acc. (%)	F1-score	Acc. (%)	F1-score
Majority	88.2	N/A	88.2	N/A
RF	90.5	.68	94.9	.85
LGBM	90.8	.71	95.6	.88
STResNet	90.0	.72	95.6	.90

4.2.3 *Affect and Mood Recognition Using Behaviour Data and Wrist-Device Data.*

Affective computing is an interdisciplinary field that develops systems that recognise, interpret, and simulate human affect [84, 114]. A fundamental assumption is that different mental states (e.g. emotions and stress), and different intensities of those states, manifest through physiological and behavioural changes. A variety of wearables can capture these changes, including those available in the Empatica device.

The affect and mood variables that we model in these experiments are the reported levels of: understanding, satisfaction, energy, stress, tiredness, peacefulness, motivation, calmness, enthusiasm, and worriedness. In addition to these low-level variables, we also calculated levels of:

- *Positive Activation*: average value calculated over the levels of enthusiasm, motivation, wakefulness, and energy.
- *Negative Activation*: average value calculated over the levels of anger, nervousness, stress, and worriedness.
- *Valence*: average value calculated over the levels of satisfaction and happiness.

Different than our previous ML experiments where the models used as input 2-minute data windows, in these experiments one input instance represents one session (e.g. a 2-hour class). That is because the self-reported data (labels) corresponds to the overall sessions. To avoid having bigger number of features than the number of training instances (which could cause an overfitting problem), we selected only a few representative features from each Empatica signal. These features were: mean skin temperature, mean EDA, number of SCR peaks, EDA sum of positive derivative, EDA average peak amplitude, top three frequencies of the accelerometer's magnitude, and the ten HRV features from the BVP data. As an additional step, we performed session-based aggregation over these features. The aggregation is needed because the eighteen features were calculated using a sliding window of two minutes with 50% overlap, but the questionaries data only corresponds to one full session (two classes with a break in between, which amounts to an overall duration of 2 hours). We used six aggregation functions: median, standard deviation, sum, 5th quantile, 95th quantile, and quantile difference (Q95 − Q5). After the aggregation, we had 671 instances with 108 Empatica-based features (18 main features x 6 aggregation functions) and five questionnaire-based behaviour features.

In addition to the Empatica features, we also used five questionnaire-based behaviour features, including: *study hours*, *daily activity*, *daily breakfast*, *daily caffeine intake*, and *daily sleep quality*. These features represent the participant's behaviour in the 24 hours prior to the recording session.

We used regression models in these experiments, given that each of the target affect and mood variables are on a Likert scale between 1 and 5. We investigated two types of regression models, models that use only questionnaire-based behaviour features (RF with B input and LGBM with B input models in Table 12), and models that use both behaviour features and Empatica features (RF with B+E input and LGBM with B+E input models in Table 12). For the best-performing models, we also explored feature normalisation, i.e. we performed person-specific standardisation for the Empatica features to decrease the inter-subject variability of the physiological data (RF with B+EN input and LGBM with B+EN input models in Table 12).

We evaluated the models using 5-fold cross-validation (without temporal shuffling), where all the data collected in one day belongs only in one of the folds. The regression models were retrained for each affect/mood label. The Dummy model in Table 12 outputs the mean value per label.

From the Mean Absolute Error (MAE) values in Table 12, we can see that the models that use only the behaviour features (RF/B Input and LGBM/B input) performed worse than the Dummy regressor for all of the thirteen labels. On the other hand, the models that used the combination of behaviour features and Empatica features (RF/BE and LGBM/BE models) slightly outperformed the Dummy regressor. The best results are achieved using the combination of behaviour features and normalised Empatica features (B + EN). For example, for the label valence (VA in Table 12), RF and LGBM achieved MAE of 0.73, which is a slight improvement compared to the MAE of 0.80 calculated for the Dummy regressor. These results indicate that the physiological data contains some information related to the participants' affective states. Nevertheless, modelling affect is a complex task that requires more advanced approaches.

We also explored person-independent models evaluated using Leave-One-Subject-out (LOSO) cross-validation. The results of these experiments are included in Appendix F.3. None of the models performed better than the

Table 12. Mean Absolute Error for affect and mood regression models using questionnaire-based behaviour data only (**B**), or behaviour data and Empatica data (B+E), or behaviour data with **N**ormalised (**S**tandardised **F**eatures) Empatica data (B+EN), or Baseline -no features- (Base (n.f.)). Each label is in the range [1, 5]. The dummy regressor outputs the mean value per label. The models were evaluated using Person-dependent 5-fold cross-validation (without temporal shuffling). Columns represent the labels as follows: **und**erstanding, **sat**isfaction, **en**ergy, **str**ess, **tir**edness, **pea**cefulness, **mot**ivation, **cal**mness, **ent**husiasm, **wor**riedness, **P**ositive **A**ffect, **N**egative **A**ffect and **VA**lence.

N	Model	Input type	und	sat	en	str	tir	pea	mot	cal	ent	wor	PA	NA	VA
No	RF	B	0.92	0.92	0.88	0.96	0.94	0.83	0.87	0.81	0.87	0.94	0.63	0.70	0.83
	LGBM	B	0.87	0.89	0.86	0.93	0.90	0.77	0.84	0.78	0.84	0.92	0.60	0.66	0.79
	RF	B+E	0.84	0.87	0.85	0.88	0.88	0.75	0.84	0.77	0.82	0.86	0.59	0.62	0.78
	LGBM	B+E	0.86	0.88	0.85	0.89	0.90	0.75	0.84	0.79	0.82	0.88	0.60	0.63	0.78
S.F.	RF	B+EN	0.79	0.82	0.87	0.85	0.92	0.70	0.84	0.77	0.79	0.83	0.59	0.55	0.73
	LGBM	B+EN	0.79	0.83	0.91	0.84	0.91	0.69	0.83	0.76	0.80	0.82	0.59	0.55	0.73
	Dummy	Base (n.f.)	0.89	0.89	0.88	0.96	0.98	0.77	0.89	0.81	0.82	0.95	0.62	0.67	0.80

baselines, confirming the complexity of the task (modelling affective states), and the need for more advanced, personalised ML approaches.

5 DISCUSSION

In this section, we will (i) discuss the limitations of our work, (ii) explain our intended use of the dataset for human memory augmentation, and (iii) conclude the discussion with a brief summary of the dataset and potential applications of relevance to the UbiComp community.

5.1 Limitations

5.1.1 Lack of Continuous Physiological Data. One of the main limitations of our dataset is the limited amount of physiological data. As we only collected participants' signals during classes, the data is not continuous and does not provide a complete understanding of users' routine behaviour context. Several reasons motivated this decision.

First, collecting physiological data outside the classroom would have required participants to use the device all day, increasing their experiment workload and the possibility of user error during device charging and data synchronisation. Maintaining a balance between the participant's perceived cost of the experiment and its reward is important to decrease dropout rates [28].

Second, our experimental design necessitated randomly assigning a device to participants for each class to enable user identification without device-specific bias. If participants had kept their devices for longer periods, the logistics of randomising the device would increase in difficulty, as participants would need to charge and synchronise the devices every day before meeting with us to sort them again.

Third, we prioritised data diversity over quantity, as we had more participants (42) than available devices (31).

Lastly, our primary interest was collecting in-class physiological signals for analysing students' attention and arousal levels due to their relationship with memory. We could not compromise on having less in-class data due to errors arising from extended out-of-class data collection.

Future work should aim to conduct a more extensive data collection that includes students' physiological signals outside class, along with additional affective self-reports throughout the day. Such an experimental design, compared to our dataset, would obtain a more comprehensive understanding of participants' contextual mood and behaviour.

5.1.2 Reliability of Self-Reports. Our dataset relies mostly on self-reports, which can be subject to different types of bias, such as self-report bias, recall bias, social desirability bias, and bias errors resulting from the wrong interpretation of questions.

Self-report bias is perhaps the most difficult to address, as it is impossible to know how accurately students can assess their feelings, moods, behaviours and thoughts. As described in the Validation (4) and some of our prior work [36, 67], we assess the validity of the self-reports by including objective data such as grades and physiological signals in our analysis.

To minimise *recall bias*, we sent survey reminder notifications right after each class ended. Our analysis shows that students answered surveys, on average, within 18 minutes of receiving the notification. We believe this time frame is short enough to avoid this type of bias.

We took several steps to avoid *social desirability bias*, i.e. the possibility of students trying to answer self-reports in a way that would please the researchers or to appear more socially desirable. First, we informed students that their self-reports would not affect their grades, and that their data would only be analysed after grading. Lastly, none of the two researchers involved in data collection was involved in grading quizzes or final exams for the two courses in question.

To address the potential *wrong interpretation of questions*, we randomly asked participants if they had any concerns about the content of the surveys during the first days of data collection. Only few students had doubts (although never about the same question), so we (i) provided them with prompt clarifications and (ii) immediately disambiguated the corresponding question in the subsequent versions of the survey.

5.2 Human Memory Augmentation Applications

While so far we focused mainly on the use of our dataset in the area of affective computing, we believe that it also has the potential to facilitate other applications, particularly in the field of Human Memory Augmentation. The richness of the dataset provides a unique research opportunity. Compared to other datasets in similar areas, ours contains students' physiological signals while attending classes, their performance across multiple examinations, and features class content in different modalities (video, audio and slides) with timestamped events (e.g. breaks) and topic descriptors as metadata.

As described in Section 2.6 in the Related Work, the analysis of physiological signals can be used to recognise students' affect and attention levels throughout classes. By adding timestamped topics to the analysis (i.e. when a particular topic was being taught during a class), we could also identify the students' attention levels during specific topics. Furthermore, as both courses in the dataset had recurrent quizzes (at least as often as every two weeks), we can also examine the students performance on a per-topic basis. This analysis may help us predict the topics a student may be struggling with.

Moreover, the multimodality of captured and available class content is also beneficial for research in the Human Memory Augmentation area. Since human memory is cue-driven, and cues can be of different types (with their efficacy also depending on its type), the class content could be used to create different sets of cues (e.g. flashcards) that the students could review based on their learning style preferences. A spaced repetition approach could be used to present these cues to the students and improve their learning.

The most interesting aspect of this research is that this whole process could be partially automated by an HMA system, relieving the user of some of the workload and assisting them with memory recall. Lectures are recorded by many universities worldwide and made available to students. Physiological signals can be automatically filtered, processed and analysed, and cues could be automatically generated and presented to the user at relevant times (e.g. by monitoring their cognitive load to optimise their receptivity to study prompts [20]). The user would only need to take care of the device (i.e. charging and synchronising) and review the content at the appropriate time for learning.

5.3 Conclusion

In this work, we presented a dataset we collected during a 13-week study in a university setting. The dataset comprises raw physiological data (namely, ACC, EDA, TEMP, BVP) and expert features (Section 3.5, Appendix A) from 42 student participants and 2 lecturers during 26 recording sessions (i.e. lectures) along with daily surveys covering the participants' habits (such as study, physical activity, sleep quality, caffeine and breakfast intakes). The dataset also includes students' performance in the different course examinations (i.e. quizzes, assignments and exams) and timestamped class topics.

Our dataset differs from similar work in the area in several aspects. First, multiple datasets in the affective computing domain only contain one session per subject, limiting the analysis of inter-subject and intra-subject correlations. Paired with the daily self-reports, this longitudinal study also enables the analysis of trends and habits throughout the semester, as well as the collective affective status of the students during class. Second, some of the other datasets were collected in laboratory or uncontrolled settings, increasing the number of artefacts in the data. Our dataset benefits from being an in-the-wild experiment in a relatively controlled environment (structured classes in a University setting), minimising the risk of spurious data. Third, compared to the most important work in a similar university setting, StudentLife [120], our dataset contains the participants' physiological signals attending to one of their courses, which expands the dataset's possibilities.

Regardless of our primary research motivation to use this dataset for exploring HMA systems, we believe that DATASET offers a significant contribution to the UbiComp community, including but not limited to affective computing tasks. We covered some of the possibilities that this dataset offers in Section 4 - Validation, using statistical analysis and ML models. We still envision much more other applications for it, such as:

- comparing physiological signals of
  - caffeine drinkers vs. not drinkers
  - breakfast eaters (small? big?) vs. no breakfast
  - physically active vs. sedentary
  - top-performers vs. low-performers
  - good sleep quality vs bad sleep quality (either in aggregated form or within-subjects)
- user identification from physiological signals
- device identification from device-specific bias
- activity detection (in-class vs breaks)
- affect prediction/modelling from physiological signals and self-reports
- engagement detection from self-reports (e.g. effects of sleep, nutrition, physical activity on class sentiment)
- learning performance prediction
- modelling behaviour (exercise level, sleep quality, food and caffeine intake)

The dataset is available to researchers after the signature of a sharing agreement (see Section 6 - Dataset and Code Availability).

6 DATASET AND CODE AVAILABILITY

The dataset is available to researchers after the signing of a sharing agreement. Please contact the first author for more information.

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APPENDIX A LIST OF FEATURES

Table 13. List of features calculated from the raw data extracted from Empatica's E4

Raw Signal	No. of features	Description of features
Inter-Subject Correlation	4	One ISC per signal (EDA, HR, ACC and ST)
PP intervals / IBI	26	•13 HRV features based on the PP intervals: ○Time domain: mean heart rate, mean PP interval, the standard deviation of the PP intervals (also known as SDNN), the standard deviation of the squared successive differences (SDSD), the root of the mean of the squared successive differences (RMSSD), the proportion of successive differences that are larger than 20 milliseconds (pnn20), the proportion of successive differences that are larger than 50 milliseconds (pnn50), Poincare plot indices (SD1 and SD2) and their ratio (SD1/SD2). ○Frequency domain: LF (total spectral power in bands between 0.04 and 0.15 Hz), HF total spectral power in bands between 0.15 and 0.4 Hz, and the ratio LF/HF. •13 HRV features based on Empatica's IBI data
Continuous Heart Rate	22	•11 statistical features of the raw signal •11 statistical features of the first derivative of the raw signal
Blood Volume Pulse	22	•11 statistical features of the raw signal •11 statistical features of the first derivative of the raw signal
Skin Temperature	22	•11 statistical features of the raw signal •11 statistical features of the first derivative of the raw signal

Table 14. List of features calculated from the raw data extracted from Empatica's E4 (continuation)

Raw Signal	No. of features	Description of features
Acceleration	50	•6 features: top 3 frequencies in the Power Spectral Density (PSD) and the 3 corresponding values •11 statistical features of the magnitude signal •11 statistical features of the first derivative of the magnitude signal •11 statistical features of the PSD •11 statistical features of the PSD's first derivative
Electrodermal Activity	63	•11 statistical features of the raw signal •11 statistical features of the first derivative of the raw signal •12 features from the raw EDA signal: mean squared amplitude (power of the signal), the sum of the positive amplitude changes, duration of the positive amplitude changes, rate of positive changes (i.e. positive amplitude changes per second), the sum of the first derivative of the tonic component, the mean value of the filtered EDA signal after subtracting the tonic component, and six features describing the power of the spectrum of the EDA signal in the frequency bands ([0-0.5] with a step of 0.1 Hz.) •19 features related to the SCRs: number of SCRs, rate of SCRs (i.e. SCRs per second), mean amplitude of all SCRs, eight features describing the biggest SCR in the given window (in Figure 4, the largest amplitude near the 20-th second) including amplitude change w.r.t. the start, amplitude change w.r.t. the end, increase duration w.r.t. the start, decrease duration w.r.t. the end, overall duration from the peak-start to the peak-end, the slope of the peak w.r.t. the start, the slope of the peak w.r.t. the end, and the ratio between the increase duration and the decrease duration. In addition to describing the biggest SCR, we also calculated the same eight features from all the SCRs in a given window and summarised them using averages, resulting in eight average values. •10 features related to periods with significant EDA changes: amplitude increase, amplitude decrease, duration of all increase and decrease segments (two features), the mean amplitude changes of the increase and the decrease segments (two features), the duration multiplied by by the amplitude changes (two features), the difference between the maximum and the minimum values on the increasing and decreasing segments (two features), and the speed of the changes (two features).

The 11 statistical descriptors are: the first four statistical moments, the mean value of the 1st. and 2nd. derivative, the 25th and 75th percentiles, the quartile deviation, the difference between the maximum and minimum values, and the coefficient of variation.

APPENDIX B INITIAL SURVEY QUESTIONS

Table 15. List of questions in the Initial Survey. Question types are **M**ultiple **S**election, **M**ultiple **C**hoice, and **F**ree **T**ext.

N°	Type	Question	Values
1	M.S.	What do you usually have for a weekday breakfast?	'Cereal/Müsli', 'Yoghurt', 'Bread', 'Eggs', 'Fruit', 'Vegetables', 'Coffee', 'Tea', 'Milk', 'Juice/Smoothie', 'Energy drinks', 'Other (please specify)'
2	M.C.	Do you exercise regularly?	0: 'No', 1: 'Yes'
2.1	M.S.	What kind of exercise do you usually do?	'Aerobic Exercise (e.g. jogging, cycling, swimming)', 'Anaerobic / Strength Conditioning (e.g. gym workout)', 'Flexibility Training (e.g. Yoga)', 'Sports (e.g. Soccer, Tennis)', 'Other (please specify)'
3	M.C.	Do you smoke?	0: 'No, I never did', 1: 'No, not anymore' 2: 'Yes', 3: 'I prefer not to say'
3.1	M.C.	How long ago did you quit?	1: 'Around 1 month ago', 2: 'Around 3 months ago', 3: 'Around 12 months ago', 4: 'More than 12 months ago'
3.2	M.C.	How much do you usually smoke?	1: 'No more than 5 cigarettes a week', 2: 'No more than 5 cigarettes a day', 3: 'Between 6-10 cigarettes a day', 4: 'More than 10 cigarettes a day', 5: 'I prefer not to say'
4	M.S.	Which languages are your mother languages?	'English', 'Italian', 'German', 'French', 'Spanish', 'Mandarin', 'Arabic', 'Not listed'
5	M.C.	How would you describe your English language level?	1: 'Basic (A1, A2)' 2: 'Intermediate (B1, B2)' 3: 'Proficient (C1, C2)' 4: 'Native speaker'
6	M.C.	How likely are you to actively seek information from external sources (material not covered in class) during your studies?	1: 'Very unlikely' 2: 'Unlikely' 3: 'Likely' 4: 'Very likely'
7	M.C.	How confident do you feel that you have sufficient academic skills and abilities for Course_0x?	1: 'Not confident at all', 2: 'Slightly confident', 3: 'Somewhat confident', 4: 'Fairly confident', 5: 'Completely confident'
8	M.C.	What kind of environment are you most likely to be studying in?	1: 'Quiet environment, alone', 2: 'Quiet environment, group' 3: 'Noisy environment, alone', 4: 'Noisy environment, group' 5: 'Other (please specify)'
8.1	F.T.	Please describe your usual study environment	
9	M.C.	What grade do you expect to achieve in Course_0x?	1: '4.0 or below' 2: 'Between 4.0 and 5.5 (included)' 3: 'Between 6.0 and 7.0 (included)' 4: 'Between 7.5 and 8.5 (included)' 5: '9.0 or more'

APPENDIX C DAILY SURVEY QUESTIONS

Table 16. List of questions in the Daily Survey. Question types are **M**ultiple **C**hoice, **L**ikert **S**cale, and **P**rompt (i.e., no question).

N°	Type	Question	Values
1	M.C.	Did you attend the Course_0x Lecture today?	0: 'No', 1: 'Yes', 2: 'We did not have class today (or: not yet)',
10.1.1	L.S.	How much of today's session have you understood?	1: 'Nothing at all', 2: 'A little', 3: 'Some of it', 4: 'A lot', 5: 'Everything'
10.1.2	P.	Please rank your feelings right after the Course_0x class with respect to the emotions specified in the following pages.	
10.1.2.1	L.S.	Satisfaction	1: 'Very satisfied', 2: 'Somewhat satisfied', 3: 'Undecided', 4: 'Somewhat dissatisfied', 5: 'Very dissatisfied'
10.1.2.2	L.S.	Energy	1: 'Full of energy', 2: 'Somewhat energized', 3: 'Undecided', 4: 'Somewhat drained', 5: 'No energy at all'
10.1.2.3	L.S.	Stress	1: 'Very stressed', 2: 'Somewhat stressed', 3: 'Undecided', 4: 'Somewhat relaxed', 5: 'Relaxed'
10.1.2.4	L.S.	Tiredness	1: 'Very tired', 2: 'Somewhat tired', 3: 'Undecided', 4: 'Somewhat awake', 5: 'Wide awake'
10.1.2.5	L.S.	Peacefulness	1: 'Very peaceful', 2: 'Somewhat peaceful', 3: 'Undecided', 4: 'Somewhat angry', 5: 'Very angry'
10.1.2.6	L.S.	Unhappiness	1: 'Very unhappy', 2: 'Somewhat unhappy', 3: 'Undecided', 4: 'Somewhat happy', 5: 'Very happy'
10.1.2.7	L.S.	Motivation	1: 'Spiritless', 2: 'Somewhat spiritless', 3: 'Undecided', 4: 'Somewhat motivated', 5: 'Highly motivated'
10.1.2.8	L.S.	Calmness	1: 'Very calm', 2: 'Somewhat calm', 3: 'Undecided', 4: 'Somewhat nervous', 5: 'Very nervous'
10.1.2.9	L.S.	Enthusiasm	1: 'Very enthusiastic', 2: 'Somewhat enthusiastic', 3: 'Undecided', 4: 'Somewhat bored', 5: 'Very bored'
10.1.2.10	L.S.	Worriedness	1: 'Very worried', 2: 'Somewhat worried', 3: 'Undecided', 4: 'Somewhat free of worry', 5: 'Free of worry'

Table 17. List of questions in the Daily Survey (continuation). Question types are **M**ultiple **C**hoice, and **L**ikert **S**cale.

N°	Type	Question	Values
10.2	M.C.	Did you study since yesterday's survey?	1: 'Yes, for more than 1 hour', 2: 'Yes, for less than 1 hour', 0: 'No, I didn't study
10.3	M.C.	Did you exercise since yesterday's survey?	1: 'Yes, for over 30 minutes', 2: 'Yes, for less than 30 minutes', 0: 'No'
10.4	M.C.	Did you have breakfast today?	1: 'Yes', 0: 'No', 2: 'I prefer not to answer'
10.5	M.C.	Did you have any caffeine (e.g. coffee, tea, energy drinks) this morning?	1: 'Yes', 0: 'No', 2: 'I prefer not to answer'
10.6	L.S.	How did you sleep last night?	1: 'Very badly', 2: 'Badly', 3: 'Normal/Fair', 4: 'Well', 5: 'Very well'
10.1.2v2.1	L.S.	What was your engagement level during this lecture?	1: 'Not engaged at all', 2: 'Slightly engaged', 3: 'Somewhat engaged', 4: 'Moderately engaged', 5: 'Extremely engaged'
10.1.2v2.2	L.S.	In your opinion, what was the engagement level of the presenter during this lecture?	1: 'Not engaged at all', 2: 'Slightly engaged', 3: 'Somewhat engaged', 4: 'Moderately engaged', 5: 'Extremely engaged'
10.1.2v2.3	L.S.	Please indicate your level of agreement with the following statement: "I lost track of the world while I was giving this presentation"	1: 'Strongly disagree', 2: 'Disagree', 3: 'Agree', 4: 'Strongly agree'
10.1.2v2.4	L.S.	Happiness	1: 'Very unhappy', 2: 'Somewhat unhappy', 3: 'Undecided', 4: 'Somewhat happy', 5: 'Very happy'
10.2.v2	M.C.	Did you study (book reading, exercise solving, work on assignments) for the Course_0x class since yesterday's survey?	1: 'Yes, for more than 1 hour', 2: 'Yes, for less than 1 hour', 0: 'No, I didn't study'

APPENDIX D FINAL SURVEY QUESTIONS

Table 18. List of questions in the Final Survey (pre-exam). Question types are **M**ultiple **S**election, **M**ultiple **C**hoice, **P**rompt (i.e., no question), and **L**ikert **S**cale.

N°	Type	Question	Values
1	M.S.	What do you usually have for a weekday breakfast?	’Cereal/Müsli’, ’Yoghurt’, ’Bread’, ’Eggs’, ’Fruit’, ’Vegetables’, ’Coffee’, ’Tea’, ’Milk’, ’Juice/Smoothie’, ’Energy drinks’, ’Other (please specify)’
2	M.C.	Did you exercise regularly in the past month?	’No’, ’Yes’
2.1	M.S.	What kind of exercise did you do?	’Aerobic Exercise (e.g. jogging, cycling, swimming)’, ’Anaerobic / Strength Conditioning (e.g. gym workout)’, ’Flexibility Training (e.g. Yoga)’, ’Sports (e.g. Soccer, Tennis)’, ’Other (please specify)’
3	M.C.	Do you smoke?	’No, I never did’ ’No, not anymore’ ’Yes’, ’I prefer not to say’
3.1	M.C.	How long ago did you quit?	’Around 1 month ago’, ’Around 3 months ago’, ’Around 12 months ago’, ’More than 12 months ago’
3.2	M.C.	How much do you usually smoke?	’No more than 5 cigarettes a week’, ’No more than 5 cigarettes a day’, ’Between 6-10 cigarettes a day’, ’More than 10 cigarettes a day’, ’I prefer not to say’
4	M.C.	In what kind of environment did you usually study for Course_0x during this semester?	’In a quiet environment, alone’ ’In a quiet environment in a group’ ’In a noisy environment, alone’ ’In a noisy environment in a group’ ’Other (please specify)’
5	M.C.	What grade do you expect to achieve in Course_0x?	’4.0 or below’ ’Between 4.0 and 5.5 (included)’ ’Between 6.0 and 7.0 (included)’ ’Between 7.5 and 8.5 (included)’ ’9.0 or more’
6	M.C.	How much information did you seek from external sources (material not covered in class) for Course_0x during this semester?	’Never’, ’Seldom’, ’Sometimes’, ’Often’
7.	P.	To what extent do you agree with the following statements?	
7.1	L.S.	I feel comfortable around other students in the class	1: ’Strongly disagree’, 2: ’Disagree’, 3: ’Agree’, 4: ’Strongly agree’
7.2	L.S.	My relationship with other students has motivated me to study more or try to achieve a better grade	1: ’Strongly disagree’, 2: ’Disagree’, 3: ’Agree’, 4: ’Strongly agree’

Table 19. List of questions in the Final Survey (pre-exam) (continuation). Question types are **L**ikert **S**cale and **P**rompt (i.e., no question).

N°	Type	Question	Values
7.3	L.S.	I feel comfortable talking with the Professor	1: 'Strongly disagree', 2: 'Disagree', 3: 'Agree', 4: 'Strongly agree'
7.4	L.S.	I was engaged with course materials (i.e., listening/assisting to lectures, doing readings, etc.)	1: 'Strongly disagree', 2: 'Disagree', 3: 'Agree', 4: 'Strongly agree'
7.5	L.S.	I have learnt how to do my coursework in an efficient manner	1: 'Strongly disagree', 2: 'Disagree', 3: 'Agree', 4: 'Strongly agree'
7.6	L.S.	Overall, I am satisfied with the learning experience of this course	1: 'Strongly disagree', 2: 'Disagree', 3: 'Agree', 4: 'Strongly agree'
8	L.S.	What is your level of understanding of the theoretical concepts of the lectures?	'Very Poor', 'Below Average', 'Average', 'Above Average', 'Excellent'
9.	P.	Please rate your level of understanding of the following theoretical topics: (full topic list available in dataset)	
9.x	L.S.	Topic Name	'Very poor', 'Below Average', 'Average', 'Above Average', 'Excellent'
10	L.S.	What is your level of understanding for the practical skills of the Labs?	'Very Poor', 'Below Average', 'Average', 'Above Average', 'Excellent'
11.	P.	Please rate your level of understanding of the following practical topics: (topic list available in dataset)	
11.x	L.S.	Topic Name	'Very Poor', 'Below Average', 'Average', 'Above Average', 'Excellent'
12,13,...	P.	The Big Five inventory (full question list available in dataset)	

Table 20. List of questions in the Final Survey (post-exam). Question types are **P**rompt (i.e., no question) and **L**ikert **S**cale.

N°	Type	Question	Values
1	P.	What was your level of preparedness for the following topics: (topic list available in dataset)	
1.x	L.S.	Topic Name	'Very Poor', 'Below Average', 'Average', 'Above Average', 'Excellent'

APPENDIX E LECTURER SURVEY QUESTIONS

Table 21. List of questions in the Lecturer's Survey. Question types are **L**ikert **S**cale, **N**umeric, and **P**rompt (i.e., no question)

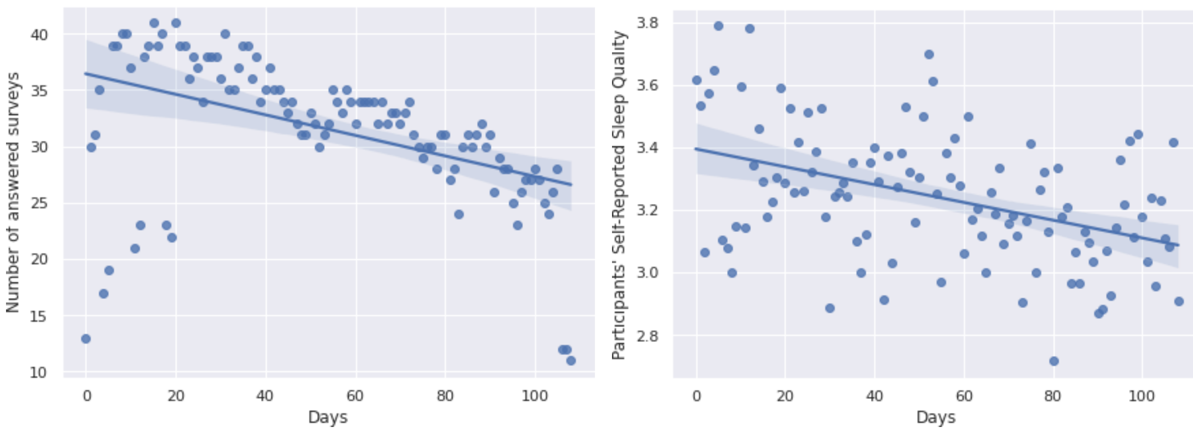
N°	Type	Text	Values
1	L.S.	What was your engagement level during your presentation?	1: 'Not engaged at all' 2: 'Slightly engaged' 3: 'Somewhat engaged' 4: 'Moderately engaged' 5: 'Extremely engaged'
2	L.S.	In your opinion, what was the engagement level of the audience during your presentation?	1: 'Not engaged at all' 2: 'Slightly engaged' 3: 'Somewhat engaged' 4: 'Moderately engaged' 5: 'Extremely engaged'
3	L.S.	Please indicate your level of agreement with the following statement: "I lost track of the world while I was giving this presentation"	1: 'Strongly disagree' 2: 'Disagree' 3: 'Agree' 4: 'Strongly agree'
4	N.	Please rate your teaching performance in today's class	1: 'Bad', 5: 'Good'
5	N.	Please rate your level of preparation for today's class	1: 'Bad', 5: 'Good'
6	P.	Please rank your feelings right after the Course_0x class with respect to the emotions specified in the following pages.	
6.1	L.S.	Worriedness, Enthusiasm, Calmness, Motivation, Happiness, Peacefulness, Tiredness, Stress, Energy, Satisfaction	Identical to "Daily Survey"

APPENDIX F ADDITIONAL ANALYSES

In this appendix, we will (i) explore some data trends throughout the semester (Appendix F.1), (ii) analyse self-reported engagement and its relationship with electrodermal activity features (Appendix F.2), and (iii) evaluate person-independent models for mood recognition (Appendix F.3).

F.1 Trends throughout the Semester

Longitudinal datasets benefit from the possibility to analyse trends throughout time. In particular, it is interesting to discover how participants' behaviour changes, and what may be the reasons for those changes. Figures 8a and 8b are an example of two different changes in participants' behaviour during the semester.



(a) Number of answered surveys by participants throughout the semester.

(b) Participants' self-reported sleep quality throughout the semester. Each dot represents the mean over all students of the self-reported sleep quality values of the day.

Fig. 8. Analysis of self-reports trends (Figure 8a, number of answered surveys; Figure 8b, self-reported sleep quality) throughout the semester.

Figure 8a shows how the number of answered daily surveys decreases over time. Sadly, attrition and participants' loss of interest [74, 102] are common problems in data collection. Even though we minimised the number of questions to avoid survey fatigue [87], some participants still expressed annoyance due to receiving notifications every day.

It is important to remark that outliers in the bottom-left section of the plot are due to a technical glitch with the survey application. During the first ~20 days of the experiment, participants from *course_01* were not able to receive surveys over the weekends, decreasing the number of answered surveys. On the other hand, outliers on the bottom right are due to *course_02* taking their final exam three days before *course_01*, thus having finished the experiment and not answering any more surveys.

Figure 8b presents the self-reported sleep quality of participants throughout the semester. Each value is the mean over all participants of the self-reported quality. Although the standard deviation is high, there is a general downward trend that implies that as the semester progresses, students feel that their sleep quality worsens.

F.2 Engagement and Skin Conductance Responses' Number of Peaks per Session

Another interesting inclusion in our dataset are the affect ground truth labels, which can be combined with the corresponding physiological data from the lectures. This pairing would enable the possibility of training models for affect prediction. Among the extracted features from the EDA signal (Section 3.5.2 - Electrodermal Activity) we have the number of SCR peaks in 2-minute data windows.

The SCR peaks represent increased physiological arousal, which previous work suggests could identify situations of momentary engagement [15]. However, these peaks may be generated by different stimuli not always related to emotional engagement [26]. Nevertheless, the presence of peaks continues to be relevant when compared to its absence, which may indicate uninteresting moments or poor motivation [6].

Figure 9 shows the analysis in our dataset of the physiological arousal, represented by the number of SCR peaks in the EDA signal, and participants' self-reported engagement.

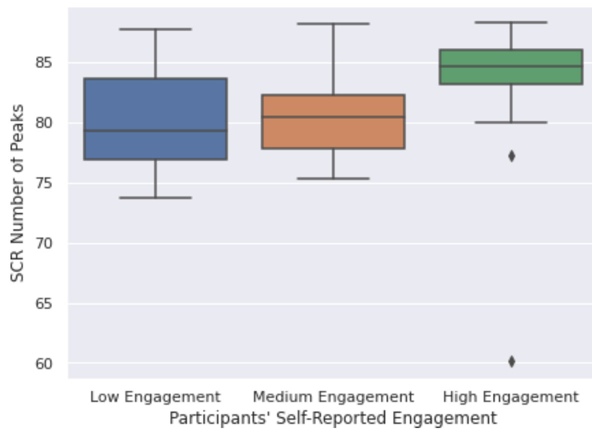


Fig. 9. Comparison of the distribution of average number of peaks over all recording sessions grouped by participants' self-reported engagement. The mean value of self-reported engagement over the whole experiment was calculated for each participant, and then participants were categorised according to that value into three bins of equal size. The SCR number of peaks is the average number of peaks per participant over all recording sessions.

Figure 9 presents participants grouped according to their answers to the question "What was your engagement level during this lecture?" (i.e. 10.1.2v2.1, *lecture_engagement*, Appendix C). The mean self-reported *lecture_engagement* value was calculated for each participant. The participants were then divided into three equal-sized groups with similar self-reported engagement levels using quantile-based discretisation with three bins. The three groups were compared with regard to the total number of peaks of each member.

The figure shows a tendency for students with high levels of self-reported engagement to also present a higher amount of peaks in their SCR signal. However, this tendency is not significant, as the Kruskal-Wallis statistical test fails to reject the null-hypothesis (with a p-value = 0.061).

F.3 Person-independent models (LOSO validation)

Table 22. Mean Absolute Error for affect and mood regression models using questionnaire-based behaviour data only (**B**), or behaviour data and Empatica data (B+E), or behaviour data with **N**ormalised (**S**tandardised **F**eatures) Empatica data (B+EN), or Baseline -no features- (Base (n.f.)). Each label is in the range [1, 5]. The dummy regressor outputs the mean value per label. The models were evaluated using LOSO cross-validation. Columns represent the labels as follows: **und**erstanding, **sat**isfaction, **en**ergy, **str**ess, **tir**edness, **pea**cefulness, **mot**ivation, **cal**mness, **ent**husiasm, **wor**riedness, **P**ositive **A**ffect, **N**egative **A**ffect and **VA**lence.

N	Model	Input type	und	sat	en	str	tir	pea	mot	cal	ent	wor	PA	NA	VA
	RF	LOSO B	0.97	0.95	0.89	1	0.94	0.85	0.89	0.84	0.88	0.98	0.63	0.75	0.86
	LGBM	LOSO B	0.92	0.92	0.87	0.97	0.91	0.79	0.87	0.81	0.86	0.96	0.62	0.71	0.82
No	RF	LOSO B+E	0.92	0.92	0.88	0.95	0.92	0.80	0.89	0.82	0.85	0.95	0.62	0.69	0.84
	LGBM	LOSO B+E	0.95	0.93	0.87	0.95	0.94	0.82	0.89	0.82	0.86	0.99	0.62	0.69	0.86
	RF	B+EN	0.90	0.94	0.91	0.97	0.97	0.80	0.90	0.85	0.84	0.96	0.63	0.70	0.84
S.F.	LGBM	B+EN	0.93	0.97	0.92	0.96	0.98	0.79	0.91	0.85	0.86	0.99	0.66	0.68	0.87
	Dummy	Base (n.f.)	0.89	0.90	0.89	0.96	0.99	0.78	0.90	0.82	0.82	0.96	0.63	0.68	0.81